

Exact colorings of graphs

OPG-37229

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Conjecture For $c \geq m \geq 1$, let $P(c, m)$ be the statement that given any exact c -coloring of the edges of a complete countably infinite graph (that is, a coloring with c colors all of which must be used at least once), there exists an exactly m -colored countably infinite complete subgraph. Then $P(c, m)$ is true if and only if $m = 1$, $m = 2$, or $c = m$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>